More NAND

Toshiba Corp. this week announced the launch of a 64GB embedded NAND flash memory module, the highest capacity yet achieved in the industry

Seagate Pulsar

Seagate announces their first enterprise grade SSD storage, based on SLC storage, the drives are available as OEM drives and are not available in retail at the present

value, it's hard to fault the second fastest drive — the SNV225-S2/64GB — it comes within 5 per cent of the Intel offering, but importantly does so at nearly half the price. We're awarding it our Digit *Best Buy*.

Hard Disk Drives

The good news for all hard drive owners — SSD is still a fair bit away from making serious inroads into HDD sales and there are perks of buying into a reliable, proven technology that has gone through numerous refinements and optimisations. What you do get is a product offering unmatchable cost per gigabyte, which is the most effective way of rating a hard drive. After all, 90 per cent of users will not be concerned with a 10 or even 30 per cent difference in performance if they have to spend 40 per cent more to reap it. WD and Seagate have different ways of classifying products, WDs method is easier to understand. In fact, we're told Seagate is in the process of making such verticals within their hard drives to clearly demarcate performance drives from value and power saving ones.

Large 2 TB hard drives have become available though the cost per GB value remains higher than any

1 TB drive. This is due to the higher areal density of the drive, where each platter is packed with more data. At the moment we're seeing platters with up to 500 GB of data.

Comparatively, platters last year had reached 333 GB. There does seem to be a stagnation of sorts happening and we're not going to see 6 and 8 TB hard drives very soon, but to be perfectly honest, there's hardly a market in India until broadband penetrates to smaller areas and what is available in metros becomes faster. Exceptions exist, and data gluts do demand more, but 2 TB is all you're going to get this year, and you pay pretty penny for owing one at the moment.

Western Digital has outdone itself when creating the 2 TB Black Edition — this drive was actually faster than a single Velociraptor, no mean task especially considering the 10,000 rpm spindle speed of the latter. In fact, in terms of speed and storage, this drive is pretty much unbeatable, if only it were priced more competitively — most users will flinch at the prospect of shelling out Rs. 18,000. Admittedly, street prices will be much lower than the mentioned MRP, but the Black Editions aren't really cheap. The WD 1 TB Black Edition seemed to have some physical fault, though we could detect no noise (which is common for damaged hard drives). The low scores meant something was really wrong, especially when the 2 TB version was blazingly fast. Unfortunately, WD couldn't rush us a replacement in time, so we tested the one piece we had. Seagate's Barracuda XT seemed interesting, until we set it up. Sadly, this drive did not do anything with the SATA 3 interface and shockingly the scores in SATA 2.0 mode were actually faster. This doesn't make sense, except that for SATA 3 mode, we were forced to use a non-Intel controller as Intel's ICH 10R does not support SATA 3.0. However, this comes as no big surprise, since given our experience, neither are SATA 2.0 HDDs really faster than their SATA 1.0 equivalents, except in benchmarks involving measuring bandwidth in burst mode. In fact with the best drives barely touching 130 MBps for read scores, which is still below the 150 MBps offered by SATA 1.0, and miles away from the 300 MBps that SATA 2.0 is supposed to deliver hard drives, in general are lagging far behind the interfaces theoretical limits. Similarly we feel it's SSDs, not HDDs that will be able to exploit SATA 3.0 when it becomes mainstream. Basing